

Task 1 — C program

```
saireddy@DESKTOP-HKQRL9G:~$ cd practical
-bash: cd: practical: No such file or directory
saireddy@DESKTOP-HKQRL9G:~$ cd OSSP
saireddy@DESKTOP-HKQRL9G:~/OSSP$ cd practical
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ pwd
/home/saireddy/OSSP/practical
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ nano fork_process.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ gcc fork_process.c -o fork_process
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ ls
fork_process  fork_process.c  practical_01  prtctical_02
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ ./fork_process
Parent process started
Parent PID : 569
Parent PPID : 483

--- PARENT PROCESS ---
Parent PID : 569
Child PID : 570
Parent waiting for child...

--- CHILD PROCESS ---
Child PID : 570
Child PPID : 569
Child is running...
Child terminating...
Child terminated.
Parent process terminating...
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ cat fork_process.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("\n--- CHILD PROCESS ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(10);

        printf("Child terminating...\n");
        exit(0);
    }
    else
    {
        // Parent process
        printf("\n--- PARENT PROCESS ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent waiting for child...\n");
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
}
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ nano process_states.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$
```

using fork())

Task 2 — Observe Process States

```
saireddy@DESKTOP-HKQRL9G:~$ cd OSSP
saireddy@DESKTOP-HKQRL9G:~/OSSP$ cd practical
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ nano process_states.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ gcc process_states.c -o process_states
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ ./process_states
Parent PID: 974
Parent PID: 974
Child PID: 975
Parent is waiting for child...
Child PID: 975
Child PPID: 974
Child is running...
Child finished sleeping.
Child terminating...
Child terminated.
Parent terminating...
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ cd practical
-bash: cd: practical: No such file or directory
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ ls
fork_process  practical_01  process_states
fork_process.c  prtctical_02  process_states.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/practical$ |
```

```

saireddy 1045 950 0 05:33 pts/4 00:00:00 grep --color=auto process_states
saireddy@DESKTOP-HKQRL9G:~$ ps -o pid,ppid,stat,cmd -p 3000,3001
PID PPID STAT CMD
saireddy@DESKTOP-HKQRL9G:~$ cat /proc/3001/status
cat: /proc/3001/status: No such file or directory
saireddy@DESKTOP-HKQRL9G:~$ top
top - 05:36:51 up 1:09, 1 user, load average: 0.06, 0.16, 0.09
tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
Mem : 3758.6 total, 3129.0 free, 466.3 used, 237.8 buff/cache
Swap: 1024.0 total, 1024.0 free, 0.0 used, 3292.3 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21524	13176	9796	S	0.0	0.3	0:13.41	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.1	0:01.43	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.1	0:00.00	init
61	root	19	-1	50368	16316	15212	S	0.0	0.4	0:12.38	systemd-journal
114	root	20	0	25216	6668	5184	S	0.0	0.2	0:13.39	systemd-udevd
123	systemd+	20	0	21344	13232	11108	S	0.0	0.3	0:05.34	systemd-resolve
124	systemd+	20	0	91036	7984	7020	S	0.0	0.2	0:11.89	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.1	0:00.34	cron
177	message+	20	0	9640	5404	4704	S	0.0	0.1	0:02.31	dbus-daemon
197	root	20	0	17980	8840	7796	S	0.0	0.2	0:00.64	systemd-logind
208	root	20	0	1756104	11036	8440	S	0.0	0.3	0:06.90	wsl-pro-service
225	syslog	20	0	222516	5976	4592	S	0.0	0.2	0:09.63	rsyslogd
228	root	20	0	3124	1972	1836	S	0.0	0.1	0:00.01	agetty
256	root	20	0	107032	23216	13764	S	0.0	0.6	0:00.81	unattended-upgr
400	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
401	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.11	Relay(403)
403	saireddy	20	0	6080	5388	3668	S	0.0	0.1	0:00.24	bash
404	root	20	0	6700	4648	3864	S	0.0	0.1	0:00.09	login
457	saireddy	20	0	20316	11512	9388	S	0.0	0.3	0:02.58	systemd
458	saireddy	20	0	21168	3584	1856	S	0.0	0.1	0:00.00	(sd-pam)
482	saireddy	20	0	6080	5340	3672	S	0.0	0.1	0:00.04	bash
871	root	20	0	3184	1120	980	S	0.0	0.0	0:00.02	SessionLeader